Chi (Bruce) Cheng

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building AI systems, agentic pipelines, and full-stack applications. Proven track record shipping production backend services, optimizing ML inference pipelines, and deploying cloud-native systems at scale. Pursuing an MS in Software Engineering at CMU; seeking Spring 2027 full-time positions or internships.

EDUCATION

Carnegie Mellon University, Silicon Valley **Dec 2026**
Master of Science, Software Engineering *Mountain View, CA*
- **Coursework:** AI in Industry, Gen AI in Practice, Machine Learning in Software Engineering, Foundation of Computer Systems, Functional Programming

University of California San Diego **Apr 2025**
Bachelor of Science, Mathematics-Computer Science
- **Coursework:** Design & Analysis of Algorithms, Advanced Data Structures, Computer Vision, Machine Learning, Recommendation Systems

EXPERIENCE

Carnegie Mellon University - Applied Generative AI (AGAI) | *Graduate Researcher* **Aug 2026 - Present**
- Developed a fact-level LLM hallucination detection pipeline using atomic fact extraction and NLI-based verification, evaluating factual consistency across TriviaQA and SQuAD benchmarks.
- Led systematic false-positive and false-negative analysis, uncovering semantic variations and verification errors to improve detection precision by 6 percentage points in benchmark testing.
- Improved hallucination detection accuracy from 82% to 90% (+8 percentage points) by refining verification criteria and optimizing evaluation workflows.

Helport AI | *AI Product Developer / Product Manager* **Sep 2024 - Jun 2025**
- Diagnosed and fixed flaws in LLM hallucination-detection experiments, using systematic evaluation and debugging to improve detection precision by 6 percentage points over the initial implementation.
- Redesigned the intent-matching pipeline, migrating from Dialogflow to a Transformer-based system using Gemini 2.0 and Vertex AI, cutting inference cost by 6x ($0.002 to $0.00031 per match).
- Built and deployed an AI dialogue-flow system with FastAPI, increasing response accuracy from 62% to 89% and reducing average handling time by 35% compared with the pre-launch baseline.
- Implemented backend APIs for real-time agent guidance, reducing development cycles from 14 days to 5 days.
- Led a two-week A/B test of a new agentic AI feature, using product analytics to increase client follow-up rates by 15% relative to the control group (from **35%** to **50%**).

Convoloo | *Software Development Engineer Intern* **Jul 2024 - Sep 2024**
- Developed an AI medical chatbot using FastAPI, LangChain, React, and Google Cloud, enabling identification of 50+ common diseases.
- Reduced average response time by 40%, from 2.5 seconds to 1.5 seconds under peak-load testing, by implementing RAG-based knowledge retrieval with Terraform and MySQL
- Led backend integration with LangServe and LangGraph using FastAPI and NestJS, streamlining service deployment.

PROJECTS

Robin - AI-Powered Personal Workspace **Jul 2026 - Present**
- *TypeScript, React, Next.js, Node.js, SQLite, Pi Agent SDK*
- * Built an AI-powered personal workspace, integrating Gmail, Google Calendar, and Telegram to turn natural-language requests and incoming emails into tracked tasks and interview schedules.
- * Designed shared domain modules consumed by REST APIs and agent tools, centralizing business rules and enforcing capability-scoped tool allowlists to restrict agent access.
- * Engineered a job-discovery and ranking pipeline combining automated collection with LLM-based evaluation, requiring supporting job-description and résumé evidence before recommending high-match roles.

NightyNight | https://nightynight-1.onrender.com/ **Apr 2025 - Jul 2026**
- Independent AI Project
- Built a full-stack AI story generator with React/TypeScript frontend and FastAPI backend over SSE for streaming.
- Designed a LangGraph multi-agent pipeline parallelizing chapter writing across LLM nodes with age-adaptive prompting.
- Integrated ElevenLabs Flash v2.5 TTS with one-shot generation (up to 40,000 chars) across 7 narrator voices.

Parking Spot Locator | https://psl.fogx.link | ***CMU / BOSCH*** **Aug 2025 - Dec 2025**
- CMU / BOSCH Research Collaborator
- Built a Bosch-sponsored locator using vision-language models and CLIP embeddings for spatial reasoning from sensor data.
- Optimized semantic search pipeline, reducing parking query latency from 45s to 15s (3x speedup).

SKILLS

- **Development & Cloud**: Python, Java, C/C++, JavaScript, TypeScript, Go, FastAPI, Flask, Node.js, Express.js, MySQL, MongoDB, SQLite, Restful API, OAuth, AWS, GCP, Docker, k8s, Terraform, GKE, CI/CD, React, Next.js, Steamlit, Figma, Git, Agile/Scrum
- **AI / ML & Frameworks**: LangGraph, LangChain, RAG, LLM APIs, OpenAI API, Vertex AI, PyTorch, TensorFlow, Hugging Face, vLLM, Milvus, Tool Calling, Agentic Workflows, milvus